DATA SCIENCE FINAL PROJECT

# Data-Driven Bank Marketing

## Term Deposit Prediction

Client Ranking for Campaign Prioritization



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# Executive Summary

| | |
|---|---|
| **Background** | Bank offers term deposits through direct marketing calls. Project focus: improve client prioritization for subscription campaigns. |
| **Problem** | Only 11.27% of observations converted, so campaign effort needs better prioritization. |
| **Stakeholder** | Marketing / Campaign Manager uses ranking output; CRM, Product, and Contact Center support execution. |
| **Goal** | Predict subscription score before contact; target = y = yes, meaning client subscribes to term deposit. |
| **Approach** | Analytics + Machine Learning Classification using Gradient Boosting. |
| **Metric** | Average Precision because positive class is imbalanced and subscriber ranking matters. |

**41,188**
historical observations

**11.27%**
positive class

**AP 0.4810**
test-set ranking

**0.60**
selective scenario

### Main decision

Use prediction score to prioritize clients before contact attempt.

# Profile Group Intro

Use case: term deposit subscription prediction for direct marketing campaign.
Primary stakeholder: Marketing / Campaign Manager.
Supporting stakeholders: CRM Team, Product Manager, Contact Center.
Team role: convert historical campaign data into client ranking support.



# Problem Formulation

### Stakeholder need

Marketing teams need to prioritize clients before contact because only 11.27% converted.

### ML problem

Predict each eligible client's subscription score during campaign execution before contact attempt.

### Output

Prediction score for client ranking and campaign prioritization.




### Why machine learning is used

Many features interact: profile, campaign context, previous response, and economic context. The model converts these patterns into a ranking score.

# Data Understanding

Source: Bank Marketing Campaign Dataset from historical direct-marketing calls.
Unit of analysis: one client observation in campaign context.
41,188 observations, 21 columns: 20 candidate features + 1 target.
Target y: yes = subscription, no = no subscription.
No external data joined for modeling; external contact-cost benchmark only supports business scenario impact.

### Client profile

age, job, marital, education

### Financial profile

default, housing, loan

### Campaign context

contact, month, day_of_week, duration, campaign

### Previous campaign

pdays, previous, poutcome

### Economic indicators

emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed

### Target

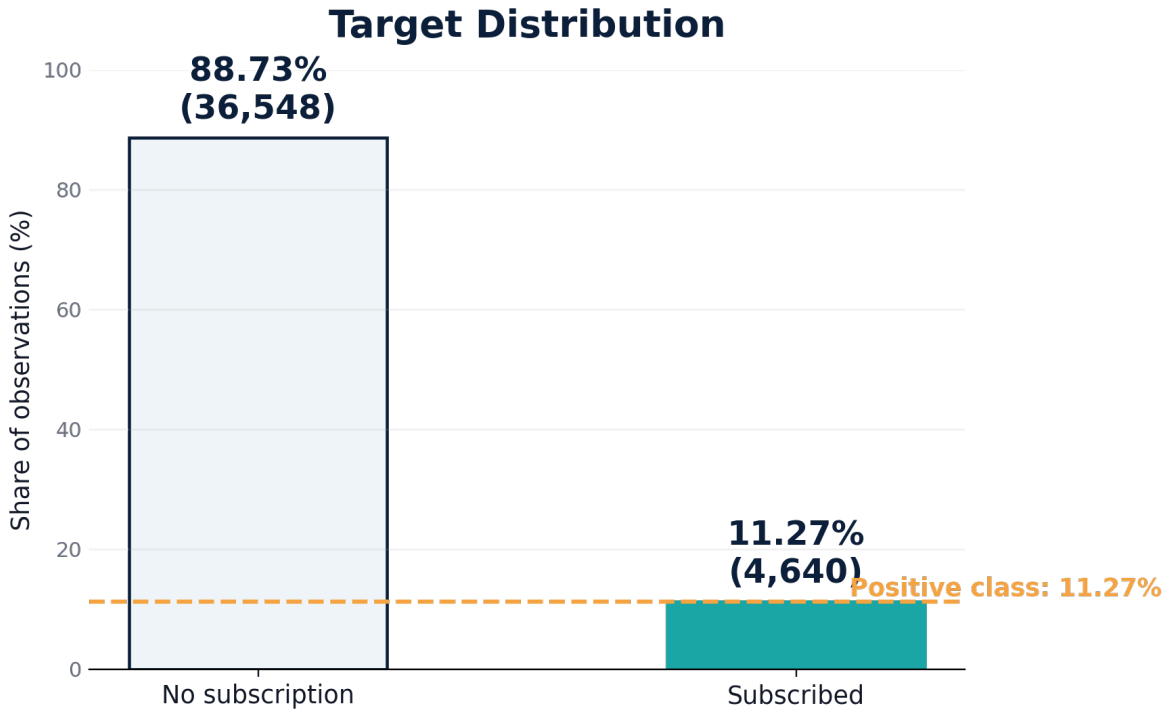
y = term deposit outcome

# Attribute Reference

| Feature group | Attributes | Business meaning |
|---|---|---|
| **Client & Financial** | age, job, marital, education; default, housing, loan | Client background and loan/default context. |
| **Campaign Context** | contact, month, day_of_week, campaign | Channel, timing, and number of current campaign contacts. |
| **Leakage Diagnostic** | duration | Call duration; excluded because it is known only after contact finishes. |
| **Previous Campaign** | pdays, previous, poutcome | Prior contact recency, prior count, and previous campaign result. |
| **Economic Indicators** | emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed | Macro-economic context during campaign period. |
| **Target** | y | Term deposit subscription outcome: yes or no. |

# Target Is Strongly Imbalanced

Positive class: 4,640 observations or 11.27% subscribed.
Majority class: 36,548 observations or 88.73% did not subscribe.
Accuracy alone can look high while failing to detect subscribers.




**Metric rationale**

Use Average Precision to focus on Precision–Recall behavior for subscribers.

# Special Values Shape the Feature Engineering

Only the most relevant distributions are shown here because they directly affect preprocessing and feature engineering.




### Distribution insight

`pdays = 999` and `poutcome = nonexistent` are structural signals, not ordinary missing values.

### Feature engineering decision

`pdays` is transformed into previous-contact information instead of treated as a normal continuous variable.

### Cleaning decision

`unknown` is retained because it can carry semantic information in historical campaign data.

# Data Cleaning & Feature Engineering




### Exclude

`duration` is post-contact leakage; not available before contact finishes.

### Engineer

`pdays = 999` means no previous contact; transformed into previous-contact information.

### Retain

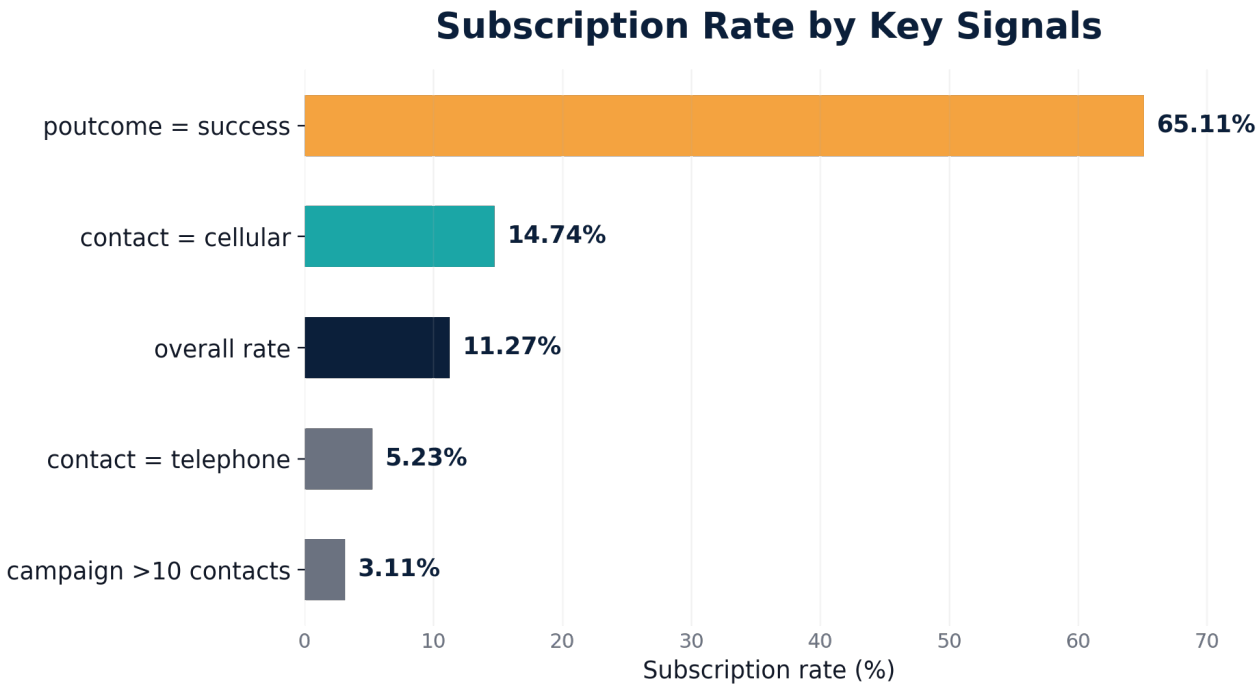
`unknown` and `nonexistent` have semantic / structural meaning; not ordinary NaN.

### Encode & Scale

OHE/Binary encoding; Robust/Standard scaler based on distribution and cardinality.

# Previous Success Is the Strongest Signal

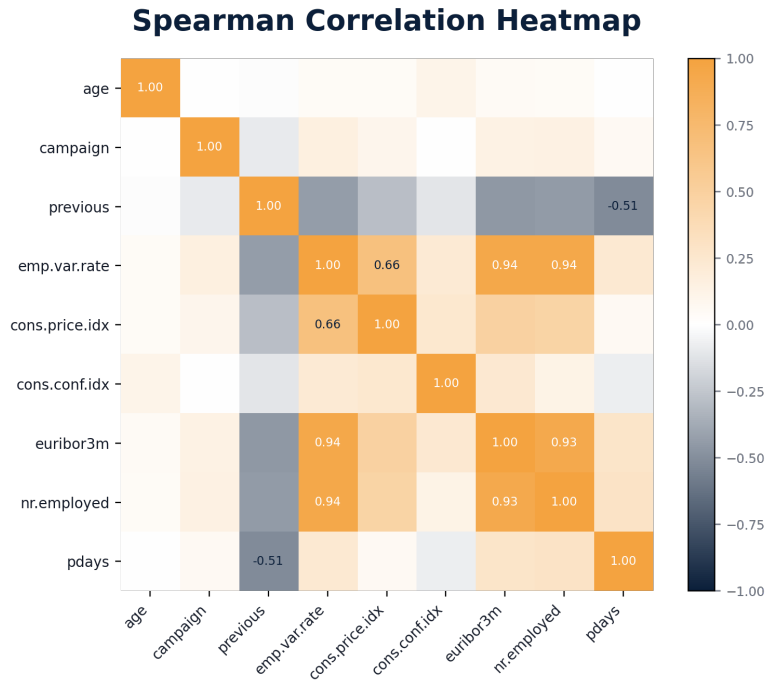
Previous campaign success has 65.11% subscription rate.
Cellular contact shows higher rate than telephone contact.
Repeated contacts do not show higher historical subscription rate.
Patterns are descriptive associations, not causal rules.




**Business cue**

Use previous response and contact history as supporting context, not standalone targeting rules.

# Economic Signals Move Together




`emp.var.rate`, `euribor3m`, and `nr.employed` are strongly correlated.

They carry overlapping information about macro conditions.

Correlation explains feature movement, not causal impact.

## Implication

Monitor economic distributions when comparing campaign performance across periods.

# Analytics + Classification Model

Analytics identifies client, campaign, and economic patterns.
Machine learning generates prediction score for prioritization.
Average Precision is used for model selection due to imbalanced target.
Threshold is treated as an operating scenario, not a universal optimum.

## Training flow




## Scoring flow




### Leakage control

Random Oversampling is applied only inside training folds; validation/test data remains unchanged.

# Gradient Boosting + ROS Has the Best CV AP

Models were compared using the same training data, preprocessing pipeline, and 5-fold Stratified Cross Validation. AP is the primary model-selection metric.




**Selected model**

Base Gradient Boosting + Random Oversampling

**Tuning result**

Tuned configuration did not improve AP.

**Decision**

Use base configuration because AP is slightly higher and simpler.

# Selected Model Generalizes to Test Data

| 0.4634 | 0.4810 | 0.8140 |
| --- | --- | --- |
| 5-fold CV AP | Untouched test AP | Untouched test ROC-AUC |

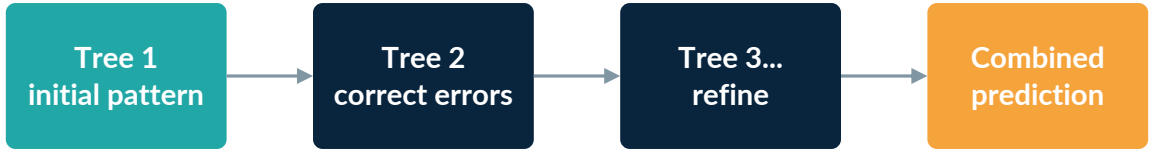
Selected model: Base Gradient Boosting + Random Oversampling.
Test-set AP is consistent with cross-validation, supporting client ranking on unseen data.
Average Precision remains the main metric because the positive class is only 11.27%.

**Metric rationale**

AP focuses on Precision–Recall behavior for subscribers under class imbalance.

# How the Final Model Works

Gradient Boosting builds decision trees sequentially.
Each tree learns remaining prediction errors from previous trees.
Random Oversampling increases minority-class exposure during training folds only.
Final model outputs a score for ranking, not an automatic business decision.



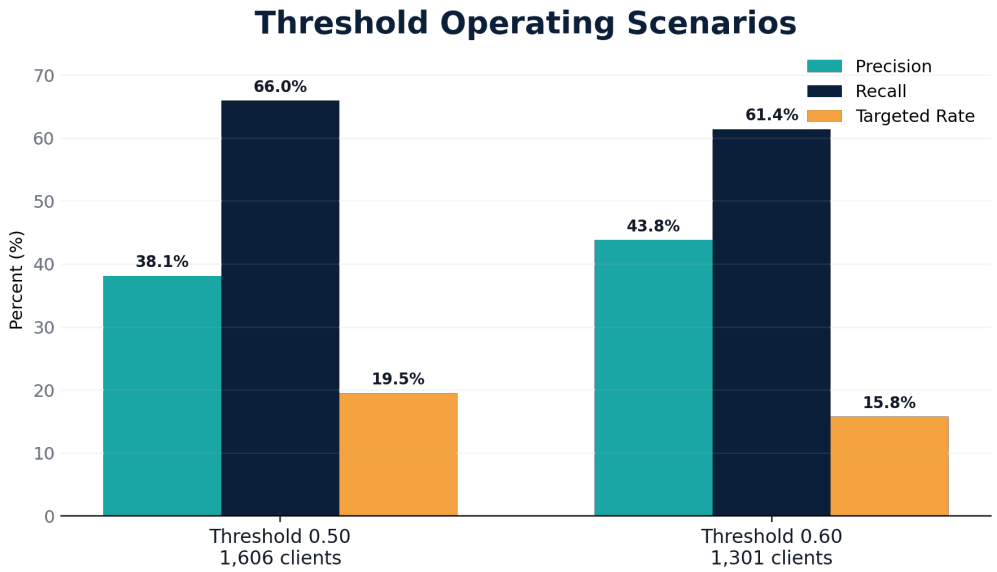

### Final configuration

n_estimators = 100
learning_rate = 0.10
max_depth = 3
random_state = 42

### Tuning decision

Tuned configuration was evaluated, but base configuration retained because AP was slightly higher and simpler.

# 0.50 Covers More; 0.60 Is More Selective




**0.50**

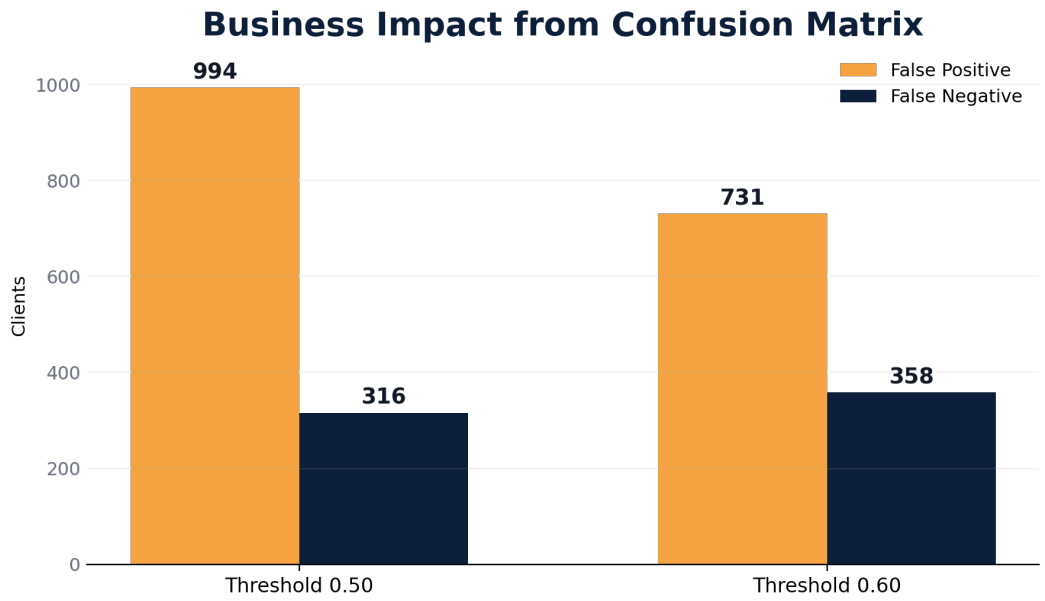
**Recall 65.95%**
**1,606 clients**

**0.60**

**Precision 43.81%**
**1,301 clients**

Threshold 0.50 captures more actual subscribers.
Threshold 0.60 reduces target pool and improves precision.
Final cut-off depends on campaign capacity and business priority.

# 0.60 Reduces 263 False Positives

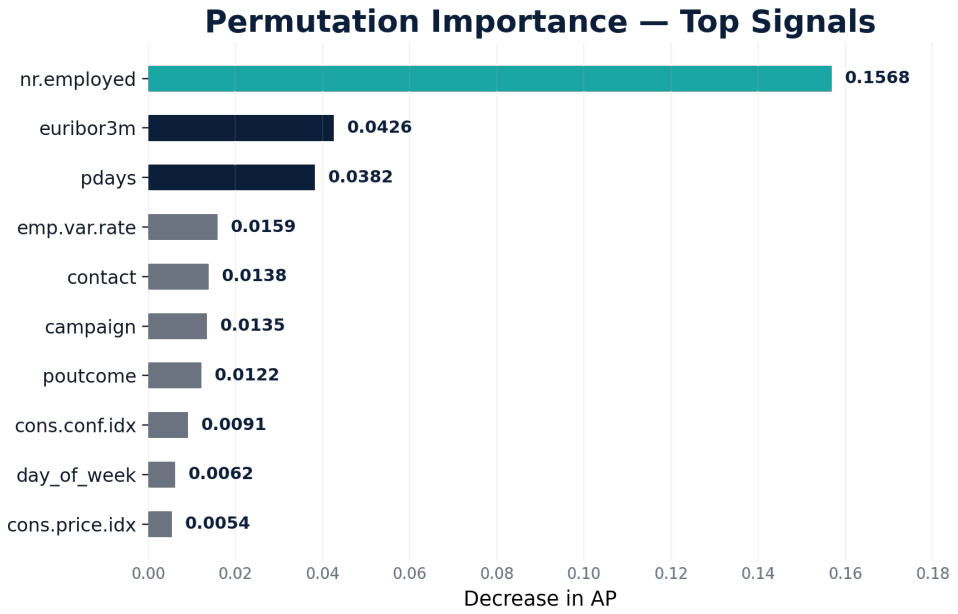



Moving from 0.50 to 0.60 reduces FP from 994 to 731.
Illustrative FP contact-cost reduction: USD 1,315 at USD 5/contact.
Sensitivity scenario: USD 2,393.30 reduction at USD 9.10/contact.
Trade-off: FN increases from 316 to 358, or 42 additional missed subscribers.

**Business impact caveat**

Actual pre-ML campaign cost baseline is unavailable. Impact is evaluated through illustrative operating scenarios.

# Economic Indicators Are Major Predictive Signals




Permutation Importance and SHAP use different definitions of importance.
Both methods consistently highlight economic indicators as major predictive signals.
`pdays` is interpreted through feature engineering as previous-contact history.
Interpretation is predictive and associational, not causal.

**Major signals**

nr.employed, euribor3m, pdays, emp.var.rate, contact, campaign, poutcome.

# Use Within Similar Campaign Context

### Temporal generalization

Dataset reflects one historical campaign context; performance may shift across periods.

### No revenue / CLV

Dataset does not contain revenue per subscription, deposit value, or customer profitability.

### No unique client ID

Rows represent campaign observations, not necessarily unique customers.

### Conditional features

`contact`, `month`, `day_of_week`, and `campaign` depend on scoring-time availability.

### Alternative models evaluated

Logistic Regression, Random Forest, Extra Trees, Gradient Boosting, class weighting, and Random Oversampling were compared.

### Potential additional data

Campaign capacity, actual contact cost, revenue per subscription, deposit amount, and customer profitability / CLV.

# What We Learned

### Diagnostic insights

Previous success is a strong historical signal.
Repeated contacts need review, not automatic escalation.
Economic conditions affect interpretation across periods.

### Predictive solution

Base Gradient Boosting + ROS selected.
Test AP 0.4810; ROC-AUC 0.8140.
Prediction score supports client ranking.

### Decision implication

0.50 prioritizes coverage.
0.60 improves selectivity.
Final threshold depends on capacity and business priorities.

### Overall conclusion

Combine historical response, campaign context, economic conditions, and prediction score for more focused campaign prioritization.

# Turn Scores into Campaign Decisions

1 **Rank eligible clients**

Score clients before contact and prioritize highest prediction score within campaign capacity.

2 **Use historical response as context**

Use `poutcome` and contact history for segmentation support, not standalone rule.

3 **Review repeated contacts**

Flag high-contact-count clients and review score before additional attempts.

4 **Monitor economic context**

Track economic feature distribution and model performance across campaign periods.

### Coverage-oriented

Threshold 0.50 • 1,606 clients
Use when campaign capacity is larger.

### Efficiency-oriented

Threshold 0.60 • 1,301 clients
Use when campaign capacity is tighter.

### Final threshold

Reflect actual campaign capacity, eligibility rules, observed performance, and business priorities.

CLOSING

# Thank You

Questions & Discussion

